Adaptive Campaign Optimization Engine

DS 223 | Marketing Analytics | Group Project

Spring 2026 | American University of Armenia

Milestone 1 | Product Roadmap + MoSCoW

Team

Role	Member
Product/Project Manager	Anna Asatryan
DB Developer	Hayk Alekyan
API/Backend Developer	Victoria Makaryan
Frontend Developer	Armine Babajanyan
Data Scientist	Davit Badalyan

Timeline Overview

Milestone	Dates	Weight	Focus
M1	Apr 1–12	25%	Problem definition, roles, roadmap, UI prototype
M2	Apr 12–14	25%	Database schema, simulation engine, LinUCB model
M3	Apr 14–May 1	25%	FastAPI backend, Streamlit frontend, Docker integration

M4	May 1–8	20%	End-to-end testing, documentation, polish
Demo	May 14	5%	Live demonstration and presentation

Milestone 1 — Foundation (Apr 1–12)

Task	Owner	Deliverable	Priority
Problem Definition document	Anna	problem_definition.pdf in repo	Must
Finalize team roles	Anna	Roles table in README.md	Must
Schedule call with Karen and Artur	Anna	Meeting booked	Must
Product roadmap + prioritized backlog	Anna	product_roadmap.pdf in repo	Must
Join GitHub Organization + repo	All	All members confirmed	Must
UI prototype / flowchart	Anna / Armine	Architecture + UI wireframe in repo	Must
Push deliverables to repo	Anna	Problem def, roadmap, UI	Must
Set up GitHub Projects board	Anna	Backlog / WIP / Review / Done	Must
Install VSCode + Project Manager ext.	All	Confirmed by all members	Must

Milestone 2 — Data + Model (Apr 12–14)

Task	Owner	Priority
PostgreSQL schema (customers, actions, rewards, model_state)	Hayk	Must
Docker Compose: db + pgadmin containers	Hayk	Must
Feature engineering	Davit	Must
Response simulation layer (logistic model, action-specific coefficients)	Davit	Must
LinUCB implementation (NumPy)	Davit	Must
Random + heuristic baseline policies	Davit	Should
ETL container: load features into DB	Victoria	Should

Milestone 3 — Backend + Frontend (Apr 14–May 1)

Task	Owner	Priority
FastAPI endpoints: POST /decide, POST /feedback, GET /metrics	Victoria	Must
SQLAlchemy models + Pydantic schemas	Victoria	Must
API container in Docker Compose	Victoria	Must
Streamlit dashboard: customer table + recommended action	Armine	Must

Cumulative reward chart (policy comparison)	Armine	Must
Simulation controls panel (n_users, alpha, cost params)	Armine	Should
Frontend container in Docker Compose	Armine	Must

Milestone 4 — Integration + Polish (May 1–8)

Task	Owner	Priority
End-to-end integration test (all containers, full loop)	All	Must
Action distribution over time chart	Armine	Should
Documentation	Anna	Should
Regret curve visualization	Davit	Could
Code cleanup, docstrings, final requirements.txt	All	Must
Thompson Sampling as alternative algorithm	Davit	Could

Demo — May 14

Task	Owner
Live demo: run simulation, show dashboard, compare policies	All

5-min presentation: problem → solution → results	Anna
Q&A preparation	All

MoSCoW Prioritization Summary

Must Have: LinUCB model, PostgreSQL, FastAPI (POST /decide, POST /feedback, GET /metrics), Streamlit dashboard (customer table, cumulative reward chart), Docker Compose, simulation engine with configurable actions, random baseline.

Should Have: Configurable alpha parameter, heuristic baseline, multiple simulation runs, action distribution chart, simulation controls panel, documentation.

Could Have: Thompson Sampling comparison, regret curve visualization, feature normalization, budget constraints on actions.

Won't Have: Deep learning, multi-step RL, real-time production deployment, full recommendation engine, distributed data infrastructure.